# Scalable Poisson Mixture Learning from Probability-Generating Function Sketches

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## Abstract

Mixtures of Poisson distributions are widely used to model heterogeneous count data, but classical likelihood-based inference can be costly on large datasets and sensitive to initialization. We study a compressed alternative based on empirical probability-generating-function sketches. The key observation is that, when the PGF of a finite mixed Poisson model is sampled along complex lines, the resulting measurements form multidimensional sums of exponentials. We exploit this structure with a joint ESPRIT recovery procedure: the empirical PGF is evaluated at *damped* query points that remain inside the closed unit disk, a global multi-snapshot subspace is estimated from several sampling directions, the per-direction rotational-invariance matrices are aligned by a common *non-unitary* (similarity) diagonalization, Poisson rates are recovered by a non-negative least-squares fit of their directional projections, and mixture weights by a simplex-constrained least-squares fit. A final refinement minimizes the exact PGF least-squares loss with a quasi-Newton method using a closed-form analytic gradient. Compared with marginal PGF–ESPRIT pipelines, the joint formulation avoids a combinatorial cross-coordinate alignment step and directly targets multidimensional components. Our experiments distinguish two questions: algorithmic comparisons under the same finite product-Poisson mixture model, and model-level comparisons involving alternative count-data models such as Poisson–Gamma variants. Because the sketch has a fixed size that is independent of the number of observations $n$, the recovery cost does not grow with $n$: empirically, the spectral fitting step stays flat while full-data EM grows linearly, and the sketch is more than two orders of magnitude smaller than the raw data at $n = 3\times10^5$. [TODO: Add final accuracy summary: PGF error, Hellinger/chi-square distances, and parameter recovery error.]

## CCS Concepts

• **Information systems** → **Data mining**; • **Mathematics of computing** → *Probabilistic algorithms*; • **Computing methodologies** → *Mixture models.*

## Keywords

data sketching, Poisson mixtures, count data, probability generating functions, ESPRIT, joint diagonalization

## 1 Introduction

Many datasets consist of nonnegative integer-valued observations: numbers of events, interactions, messages, visits, reads, or molecular counts. Finite mixtures of Poisson distributions provide a simple and interpretable model for such data. They describe heterogeneous populations through a finite number of latent components, each associated with a Poisson rate vector and a mixture weight. Estimating these parameters is a classical latent-variable inference problem, most commonly addressed through maximum-likelihood methods such as Expectation–Maximization (EM) [4, 7]. Although effective, EM-type procedures remain iterative, can be sensitive to initialization, and require repeated passes over the data or over sufficient statistics whose size grows with the model and data dimension.

Sketching offers an alternative route for large-scale inference. Instead of fitting the model directly from all observations, the dataset is first compressed into a fixed-size collection of nonlinear measurements, called a sketch, and estimation is performed from this compact representation [3, 8]. For count data, a natural choice is the probability generating function (PGF). Indeed, if $X$ is a Poisson random variable with rate $\lambda$, then its PGF is given by

$$G_X(t) = \mathbb{E}[t^X] = \exp(\lambda(t-1)).$$

Consequently, the PGF of a finite Poisson mixture is a weighted sum of exponentials. This algebraic structure suggests that mixture estimation from empirical PGF samples can be viewed as a harmonic retrieval problem, opening the door to spectral methods such as ESPRIT [9].

In this paper, we study a multidimensional sketching approach for Poisson mixture estimation based on *joint ESPRIT*. The method evaluates the empirical PGF along random-base complex lines, stacks the resulting multi-snapshot measurements, and estimates a global signal subspace shared across directions. The Poisson rate vectors are then recovered through jointly diagonalized rotational invariance matrices, while the mixture weights are estimated by solving a linear inverse problem. Compared with coordinate-wise spectral recovery, the joint formulation is designed to avoid a combinatorial alignment step between independently estimated marginal components.

**Contributions.** The contributions of this work are as follows:

- We formulate finite Poisson mixture estimation from empirical PGF sketches as a multidimensional harmonic retrieval problem, and introduce a *damped* query construction that keeps every evaluation point inside the closed unit disk while preserving the exact sum-of-exponentials structure, controlling the variance of the empirical PGF.
- We adapt a random-base joint ESPRIT pipeline to recover multidimensional Poisson rate vectors from a fixed-size PGF sketch, avoiding independent coordinate-wise pairing. The pipeline uses a non-unitary similarity diagonalization for component alignment, non-negative least squares for rates, simplex-constrained least squares for weights, and an analytic-gradient quasi-Newton refinement of the exact PGF loss.
- We evaluate the proposed approach under two distinct settings: algorithmic comparison within the finite product-Poisson mixture model, and model-level comparison with alternative count-data models, and we quantify the storage and runtime savings of the fixed-size sketch as $n$ grows. [TODO: Add final accuracy findings.]

## 2 Model and PGF Sketch

Let $X \in \mathbb{N}^d$ denote an observed count vector. We consider the finite product-Poisson mixture

$$Z \sim \sum_{k=1}^{K} \pi_k \delta_{\lambda_k}, \qquad X \mid Z = \lambda_k \sim \bigotimes_{q=1}^{d} \mathrm{Poisson}(\lambda_{qk}), \tag{1}$$

where $\lambda_k \in \mathbb{R}_+^d$, $\pi_k \geq 0$, and $\sum_{k=1}^K \pi_k = 1$. We write $\Lambda = [\lambda_1, \ldots, \lambda_K] \in \mathbb{R}_+^{d\times K}$ and $\pi = (\pi_1, \ldots, \pi_K)^\top$.

The probability generating function of $X$ is

$$G_X(t) = \mathbb{E}\left[\prod_{q=1}^{d} t_q^{X_q}\right] = \sum_{k=1}^{K} \pi_k \exp(\langle \lambda_k, t - \mathbf{1}\rangle), \qquad t \in \mathbb{C}^d. \tag{2}$$

Given independent samples $x^{(1)}, \ldots, x^{(n)}$, we use the empirical PGF

$$\widehat{G}_X(t) = \frac{1}{n}\sum_{i=1}^{n}\prod_{q=1}^{d} t_q^{x_q^{(i)}} = \frac{1}{n}\sum_{i=1}^{n} \exp\Big(\langle x^{(i)}, \log(t)\rangle\Big), \tag{3}$$

where the logarithm is applied coordinate-wise. Since the exponents $x_q^{(i)}$ are integers, this coincides with the polynomial definition of the PGF whenever the coordinates of $t$ are nonzero.

*Random-base line sampling.* We choose $M$ probing directions

$$u_1, \ldots, u_M \in \mathbb{R}^d, \quad U = [u_1^\top; \ldots; u_M^\top] \in \mathbb{R}^{M\times d},$$

with $\mathrm{rank}(U) = d$. We also draw $S$ random base points $p_1, \ldots, p_S \in \mathbb{R}^d$, shared across all directions, and sample the lines

$$\xi_{m,s,\ell} = p_s + \ell u_m, \quad m = 1, \ldots, M, \quad s = 1, \ldots, S, \quad \ell = 0, \ldots, N_m - 1. \tag{4}$$

The PGF is evaluated at the *damped* complex query points

$$t_{m,s,\ell} = r\,\mathbf{1} + \mathrm{i}\,\delta_{\mathrm{eff}}\,\xi_{m,s,\ell}, \qquad |t_{m,s,\ell,q}| \leq 1 \ \ \forall q, \tag{5}$$

where $r \in (0, 1]$ is a real damping radius and $\delta_{\mathrm{eff}} > 0$ an effective sketching scale, both derived below from a single nominal scale $\Delta > 0$. The undamped choice $t = \mathbf{1} + \mathrm{i}\Delta\xi$ used in marginal pipelines corresponds to $r = 1$, $\delta_{\mathrm{eff}} = \Delta$; it places the query points *outside* the unit disk, since $|1 + \mathrm{i}\Delta\xi_q| = \sqrt{1 + \Delta^2\xi_q^2} > 1$. The empirical PGF terms $\prod_q t_q^{x_q^{(i)}}$ then have modulus $\prod_q |t_q|^{x_q^{(i)}}$ that grows geometrically in the counts, so $\widehat{G}_X$ has heavy-tailed variance on overdispersed data. Damping the real part to $r < 1$ bounds each factor by one.

*Damped query construction.* Let $L_{\mathrm{data}}$ be a robust upper proxy for the per-observation total count, taken as the empirical 0.95 quantile of $\sum_q x_q^{(i)}$. We target a damping radius

$$r_\star = 1 - \frac{\log n}{2\,L_{\mathrm{data}}}, \tag{6}$$

chosen so that the multiplicative damping at the typical largest exponent matches the statistical fluctuation of the empirical PGF: $r_\star^{L_{\mathrm{data}}} \approx \exp(-\frac{1}{2}\log n) = n^{-1/2}$, the standard deviation scale of $\widehat{G}_X$. A smaller $r$ over-damps (bias), a larger $r$ under-controls the variance; (6) balances the two.

Feasibility then fixes the imaginary scale. At a query point, $|t_{m,s,\ell,q}|^2 = r^2 + \delta_{\mathrm{eff}}^2\xi_{m,s,\ell,q}^2$, so $|t| \leq 1$ requires $\delta_{\mathrm{eff}}\ \max_{m,s,\ell,q}|\xi_{m,s,\ell,q}| \leq \sqrt{1 - r^2}$. Writing $b_\star = \sqrt{1 - r_\star^2}$ for the imaginary budget at the target radius and $b_0 = \Delta \max|\xi|$ for the spread at the nominal scale, we rescale

$$\delta_{\mathrm{eff}} = \alpha\,\Delta, \quad \alpha = \min\Big(1, \tfrac{b_\star}{b_0}\Big), \quad r = \sqrt{1 - \min(b_0, b_\star)^2} \ \geq\ r_\star. \tag{7}$$

Thus when the nominal scale already keeps the points well inside the disk ($b_0 \leq b_\star$) no damping is needed ($\alpha = 1$, $\delta_{\mathrm{eff}} = \Delta$); otherwise the imaginary spread is clamped to $b_\star$ and the radius to $r_\star$. By construction $|t_{m,s,\ell,q}| \leq 1$ everywhere.

At the population level, the damped queries still give an *exact* sum of exponentials. Since $t_{m,s,\ell} - \mathbf{1} = (r-1)\mathbf{1} + \mathrm{i}\delta_{\mathrm{eff}}\,\xi_{m,s,\ell}$,

$$\begin{aligned} G_X(t_{m,s,\ell}) &= \sum_{k=1}^{K} \pi_k \exp\Big((r-1)\langle\lambda_k, \mathbf{1}\rangle + \mathrm{i}\delta_{\mathrm{eff}}\langle\lambda_k, p_s + \ell u_m\rangle\Big) \\ &= \sum_{k=1}^{K} c_{k,s}\,\rho_{m,k}^{\ell}, \end{aligned} \tag{8}$$

with

$$c_{k,s} = \pi_k \underbrace{e^{(r-1)\langle\lambda_k,\mathbf{1}\rangle}}_{\text{real damping}} e^{\mathrm{i}\delta_{\mathrm{eff}}\langle\lambda_k, p_s\rangle}, \ \ \rho_{m,k} = e^{\mathrm{i}\phi_{m,k}}, \ \ \phi_{m,k} = \delta_{\mathrm{eff}}\langle\lambda_k, u_m\rangle. \tag{9}$$

The key structural point is that $|\rho_{m,k}| = 1$: the snapshot index $\ell$ enters *only* through the imaginary increment $\mathrm{i}\delta_{\mathrm{eff}}\,\ell\,u_m$, while the real damping $(r-1)\langle\lambda_k, \mathbf{1}\rangle$ is constant along each line and is absorbed into the coefficient $c_{k,s}$. Consequently damping leaves the ESPRIT nodes on the unit circle and changes neither the subspace nor the diagonalization algebra below; it only rescales the recovery scale $\Delta \to \delta_{\mathrm{eff}}$. The PGF sketch is thus not a linearization: for the query family (5) the exponential model is exact, up to empirical PGF noise.

The empirical observations are

$$z_{m,s}[\ell] := \widehat{G}_X(t_{m,s,\ell}) = \sum_{k=1}^{K} c_{k,s}\rho_{m,k}^{\ell} + e_{m,s}[\ell], \tag{10}$$

where $e_{m,s}[\ell] = \widehat{G}_X(t_{m,s,\ell}) - G_X(t_{m,s,\ell})$ is the finite-sample sketching error.

*Multi-snapshot matrix model.* For each direction $m$, stack the $S$ line measurements into

$$Z_m = \begin{bmatrix} z_{m,1}[0] & \cdots & z_{m,S}[0] \\ \vdots & & \vdots \\ z_{m,1}[N_m - 1] & \cdots & z_{m,S}[N_m - 1] \end{bmatrix} \in \mathbb{C}^{N_m\times S}.$$

Define

$$v_m(\phi) = [1, e^{\mathrm{i}\phi}, \ldots, e^{\mathrm{i}(N_m-1)\phi}]^\top, \qquad V_m = [v_m(\phi_{m,1}), \ldots, v_m(\phi_{m,K})],$$

and

$$C = (c_{k,s})_{k,s} \in \mathbb{C}^{K\times S}.$$

Then

$$Z_m = V_m C + E_m. \tag{11}$$

Equivalently, stacking all directions gives the global multi-directional model

$$Z_{\text{glob}} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_M \end{bmatrix} = \underbrace{\begin{bmatrix} V_1 \\ \vdots \\ V_M \end{bmatrix}}_{V_{\text{glob}}} C + E_{\text{glob}}. \tag{12}$$

The use of shared random base points is important: it gives a common right factor $C$ across all directions, and therefore a common component basis for the joint ESPRIT step.

We assume throughout that $N_m \geq K+1$, $S \geq K$, that the projected phases $\phi_{m,k}$ are distinct modulo $2\pi$ for each direction used in ESPRIT and that $C$ has full row rank. For random base points, this rank condition holds generically provided the components are distinct and the number of snapshots is large enough.

## 3 Joint ESPRIT Recovery

The recovery follows the multi-directional ESPRIT structure suggested by (12). The goal is to estimate the projected phases $\phi_{m,k} = \delta_{\text{eff}}\langle \lambda_k, u_m \rangle$ with a consistent component index $k$ across all directions.

*Global subspace estimation.* We compute a rank-$K$ left signal subspace from the global sketch matrix (12). For instance, let

$$\widehat{R} = \frac{1}{S} Z_{\text{glob}} Z_{\text{glob}}^* \tag{13}$$

and let $\widehat{U}_s \in \mathbb{C}^{(\sum_m N_m)\times K}$ contain the $K$ dominant eigenvectors of $\widehat{R}$. In the noiseless case, there exists an invertible matrix $T_0 \in \mathbb{C}^{K\times K}$ such that

$$\widehat{U}_s = V_{\text{glob}} T_0. \tag{14}$$

Let $\widehat{U}_{s,m} \in \mathbb{C}^{N_m \times K}$ denote the block of $\widehat{U}_s$ corresponding to direction $m$.

*Rotational invariance per direction.* Let $J_m^{(1)} \in \mathbb{R}^{(N_m-1)\times N_m}$ remove the last row and $J_m^{(2)} \in \mathbb{R}^{(N_m-1)\times N_m}$ remove the first row. In the noiseless model,

$$J_m^{(2)} V_m = J_m^{(1)} V_m \operatorname{diag}(\rho_{m,1}, \ldots, \rho_{m,K}). \tag{15}$$

Since $\widehat{U}_{s,m}$ spans the same column space as $V_m$, we estimate a rotational invariance matrix by weighted least squares:

$$\widehat{\Phi}_m \in \arg\min_{\Phi \in \mathbb{C}^{K\times K}} \left\| W_m \left( J_m^{(2)} \widehat{U}_{s,m} - J_m^{(1)} \widehat{U}_{s,m} \Phi \right) \right\|_F^2. \tag{16}$$

Here $W_m$ is an optional row-weighting matrix. In the simplest implementation $W_m = I$. In practice, $W_m$ can down-weight PGF evaluations with high empirical variance or unstable numerical magnitude. This plays the same stabilizing role as row weighting in PSF-aware ESPRIT, without requiring any inversion of small factors.

In the noiseless case,

$$\Phi_m = T_0^{-1} \operatorname{diag}(\rho_{m,1}, \ldots, \rho_{m,K}) T_0, \tag{17}$$

so the matrices $\Phi_1, \ldots, \Phi_M$ are simultaneously diagonalizable and share the same eigenvectors.

*Similarity joint diagonalization and component alignment.* By (17) the matrices $\widehat{\Phi}_1, \ldots, \widehat{\Phi}_M$ share a common eigenbasis, but this basis is the columns of $T_0$, an *arbitrary invertible* matrix, not an orthogonal one. A unitary joint-diagonalization sweep, which restricts the change of basis to be orthogonal, is therefore not guaranteed to diagonalize them. We instead estimate the common change of basis by a *non-unitary* (similarity) joint diagonalization:

$$\widehat{T} \in \arg\min_{T \in \mathrm{GL}(K)} \sum_{m=1}^{M} \left\| \operatorname{offdiag}\left( T \widehat{\Phi}_m T^{-1} \right) \right\|_F^2. \tag{18}$$

The solver exploits the fact that almost any linear combination of simultaneously diagonalizable matrices has the same eigenvectors. We draw random complex coefficients $\gamma^{(j)} \in \mathbb{C}^M$, form the combination $\Phi(\gamma^{(j)}) = \sum_m \gamma_m^{(j)} \widehat{\Phi}_m$, take its eigenvector matrix $V^{(j)}$ as a candidate basis $T = \left(V^{(j)}\right)^{-1}$, and score it by the off-diagonal energy ratio

$$\eta(T) = \frac{\sum_m \left\| \operatorname{offdiag}(T \widehat{\Phi}_m T^{-1}) \right\|_F^2}{\sum_m \left\| \operatorname{diag}(T \widehat{\Phi}_m T^{-1}) \right\|_F^2}. \tag{19}$$

Over a small fixed number of trials (the deterministic all-ones combination plus 63 random draws) we keep the basis with the smallest $\eta$, and we record $\operatorname{cond}(V^{(j)})$ as a stability diagnostic: a near-singular candidate basis signals poorly separated components. This replaces the unitary Jacobi sweep of [2], which assumes an orthogonal diagonalizer that the ESPRIT matrices do not in general admit. The jointly aligned node estimates are then obtained as

$$\widehat{\rho}_{m,k} = \left( \widehat{T} \widehat{\Phi}_m \widehat{T}^{-1} \right)_{kk}, \qquad \widehat{\phi}_{m,k} = \arg(\widehat{\rho}_{m,k}). \tag{20}$$

The common diagonalizer fixes the component ordering across directions, avoiding a separate combinatorial pairing step.

*Rate reconstruction by non-negative least squares.* For each component $k$, the recovered phases satisfy

$$\widehat{\phi}_{m,k} \approx \delta_{\text{eff}} \langle \lambda_k, u_m \rangle \pmod{2\pi}, \tag{21}$$

which uses the effective scale $\delta_{\text{eff}}$ of the damped queries (9). The rate vector is the solution of the small wrapped least-squares problem

$$\widehat{\lambda}_k \in \arg\min_{\lambda \in \mathbb{R}_+^d,\ q \in \mathbb{Z}^M} \left\| \widehat{\phi}_k - \delta_{\text{eff}}\, U\lambda - 2\pi q \right\|_2^2, \tag{22}$$

where $\widehat{\phi}_k = [\widehat{\phi}_{1,k}, \ldots, \widehat{\phi}_{M,k}]^\top$. In the regime targeted by the damped construction the phases stay well inside $(-\pi, \pi)$,

$$|\delta_{\text{eff}} \langle \lambda_k, u_m \rangle| < \pi \qquad \text{for all } k, m,$$

so we set $q = 0$. In practice we first divide out the scale to form the directional projections $g_{m,k} = \widehat{\phi}_{m,k} / \delta_{\text{eff}} \approx \langle \lambda_k, u_m \rangle$ and recover each rate by the *non-negative* least-squares problem

$$\widehat{\lambda}_k = \arg\min_{\lambda \in \mathbb{R}_+^d} \|U\lambda - g_k\|_2^2, \qquad g_k = [g_{1,k}, \ldots, g_{M,k}]^\top, \tag{23}$$

solved by an active-set NNLS routine, which enforces $\lambda \geq 0$ exactly rather than clipping a sign-indefinite solution. The rank condition $\operatorname{rank}(U) = d$ ensures identifiability of the rates from their directional projections. A regular sampling step $h_m$ along direction $m$ simply replaces $\delta_{\text{eff}}$ by $\delta_{\text{eff}}\, h_m$ in $g_{m,k}$. We flag a phase-wrapping risk when $\max_{m,k} |\widehat{\phi}_{m,k}|$ approaches $\pi$ within a safety margin, since the $q = 0$ assumption is then unsafe.

*Mixture-weight estimation.* Once $\widehat{\Lambda}$ has been recovered, the mixture weights are estimated from all available PGF samples by a global simplex-constrained least-squares fit:

$$\widehat{\pi} \in \arg\min_{p\in\Delta_K} \sum_{m=1}^{M}\sum_{s=1}^{S}\sum_{\ell=0}^{N_m-1} \alpha_{m,s,\ell} \left| \widehat{G}_X(t_{m,s,\ell}) - \sum_{k=1}^{K} p_k\, e^{\langle \widehat{\lambda}_k,\, t_{m,s,\ell}-\mathbf{1}\rangle} \right|^2, \tag{24}$$

where the model term reuses the damped query points, $\langle \widehat{\lambda}_k, t_{m,s,\ell} - \mathbf{1}\rangle = (r-1)\langle \widehat{\lambda}_k, \mathbf{1}\rangle + \mathrm{i}\delta_{\mathrm{eff}}\langle \widehat{\lambda}_k, p_s + \ell u_m\rangle$, and the feasible set is the probability simplex

$$\Delta_K = \{p \in \mathbb{R}_+^K : \mathbf{1}^\top p = 1\}.$$

This is a convex quadratic program in $p$; we solve it with a sequential quadratic program (SLSQP) using the analytic gradient $2\,\Re\big(A^*(Ap-y)\big)$, where $A_{j,k} = e^{\langle \widehat{\lambda}_k, t_j - \mathbf{1}\rangle}$ stacks the model over all query indices $j = (m,s,\ell)$ and $y_j = \widehat{G}_X(t_j)$, followed by a Euclidean projection onto $\Delta_K$. The weights $\alpha_{m,s,\ell} \geq 0$ may be set to one, chosen from empirical PGF variance estimates, or used to discard numerically unstable query points.

*Refinement of the exact PGF loss.* The recovery above is algebraic and does not directly minimize a fit criterion. We therefore refine $(\widehat{\Lambda}, \widehat{\pi})$ by minimizing the exact PGF least-squares loss over all $J = \sum_m N_m S$ query points $t_1, \ldots, t_J$ (we write $y_j = \widehat{G}_X(t_j)$):

$$F(\Lambda, \pi) = \frac{1}{2J}\sum_{j=1}^{J}\left| \sum_{k=1}^{K} \pi_k\, e^{\langle \lambda_k,\, t_j - \mathbf{1}\rangle} - y_j \right|^2. \tag{25}$$

This is the same objective targeted by sketched-mixture learning with preconditioned gradient descent (PGD) [6]; we instead minimize it with a limited-memory quasi-Newton method (L-BFGS-B) supplied with a *closed-form analytic gradient*, which uses the curvature of the loss and converges in far fewer iterations, giving an accurate fit directly after the spectral step.

To enforce positivity and the simplex constraint without projections inside the optimizer, we reparametrize into unconstrained variables

$$\lambda_{qk} = e^{\theta_{qk}}, \qquad \pi = \mathrm{softmax}(\eta), \quad \pi_k = \frac{e^{\eta_k}}{\sum_{k'} e^{\eta_{k'}}}, \tag{26}$$

with box bounds on $\theta$ (rates clipped to $[\lambda_{\min}, \lambda_{\max}]$) and free logits $\eta$. Let $B \in \mathbb{C}^{J\times d}$ stack the rows $B_{j,:} = t_j - \mathbf{1}$, let $A = \exp(B\Lambda) \in \mathbb{C}^{J\times K}$ so that $A_{jk} = e^{\langle \lambda_k, t_j - \mathbf{1}\rangle}$, and let $\rho = A\pi - y \in \mathbb{C}^J$ be the residual. The gradient of (25) with respect to the natural parameters is

$$\nabla_\pi F = \tfrac{1}{J}\,\Re\big(A^*\rho\big) \in \mathbb{R}^K, \tag{27}$$

$$\nabla_{\lambda_{:k}} F = \tfrac{\pi_k}{J}\,\Re\big(B^*(\overline{A_{:k}} \odot \rho)\big) \in \mathbb{R}^d. \tag{28}$$

Propagating through the reparametrization (26) gives the gradient in the optimized variables,

$$\nabla_\theta F = \nabla_\lambda F \odot \lambda, \qquad \nabla_\eta F = \pi \odot \big(\nabla_\pi F - \langle \nabla_\pi F, \pi\rangle \mathbf{1}\big), \tag{29}$$

where $\nabla_\theta F = \nabla_\lambda F \odot \lambda$ follows from $\partial\lambda/\partial\theta = \lambda$, and the logit gradient uses the softmax Jacobian $\mathrm{diag}(\pi) - \pi\pi^\top$. The loss (25) and the exact gradient (27)–(29) are passed together to L-BFGS-B, so no finite differences are needed. In the codebase this refinement is the entry point that previously ran PGD, now backed by the analytic-gradient solver.

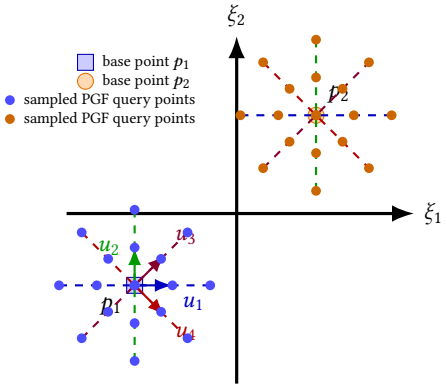



**Figure 1: Two-dimensional illustration of the random-base line sampling strategy. Two base points $p_1$ and $p_2$ are chosen randomly. Around each base point, four directions are probed, and five query locations are sampled along each direction. For visual clarity, the line samples are drawn symmetrically around the base point; the algorithm only requires consecutive samples along each direction.**

## 4 Experiments

The experiments evaluate whether PGF–JESPRIT provides an accurate and scalable estimator for finite Poisson mixtures. We explicitly distinguish algorithmic comparisons from model-level comparisons. In the algorithmic setting, all methods target the same finite product-Poisson mixture model in (1). In the model-level setting, we also report results for alternative count-data models, such as Poisson–Gamma variants, but these are not interpreted as direct algorithmic baselines for PGF–JESPRIT.

*Synthetic mixtures.* We generate $n$ independent samples from the finite Poisson mixture model in (1). The protocol varies the number of components $K$, the dimension $d$, the sample size $n$, the separation between rate vectors, and the dynamic range of the Poisson rates. We consider three representative regimes:

- *Separated components*, where the rate vectors are well separated and parameter recovery should be feasible;
- *Overlapping components*, where several rate vectors are close and the inverse problem becomes harder;
- *Under-specified fitting*, where the fitted number of components is smaller than the true one, to test robustness under model mismatch.

[TODO: Insert the final experimental grid, e.g., $K \in \{2, 3, 5, 10\}$, $d \in \{1, 3, 10\}$, and $n \in \{10^3, 10^4, 10^5\}$.]

*Compared methods.* For algorithmic comparison, the main comparator is EM for the finite product-Poisson mixture model in (1). This comparison keeps the probabilistic model fixed and contrasts likelihood-based inference with the proposed PGF-sketching and spectral-recovery pipeline. We report the proposed estimator both directly after spectral recovery (*PGF–JESPRIT*) and after the analytic-gradient L-BFGS refinement of the exact PGF loss (25) (*PGF–JESPRIT + L-BFGS*). We also report runtime and storage to assess the benefit of replacing repeated likelihood-based fitting by a

**Table 1: Synthetic algorithmic comparison under the finite product-Poisson mixture model. Errors are reported as mean ± standard deviation over independent trials; "–" marks values pending the final runs. [TODO: Fill from the synthetic experiments.]**

| Regime | Method | Rate err. | Weight err. | PGF err. |
|---|---|---|---|---|
| Separated | Poisson EM | – | – | – |
| Separated | PGF–JESPRIT | – | – | – |
| Separated | + L-BFGS | – | – | – |
| Overlap | Poisson EM | – | – | – |
| Overlap | PGF–JESPRIT | – | – | – |
| Overlap | + L-BFGS | – | – | – |
| Underfit | Poisson EM | – | – | – |
| Underfit | PGF–JESPRIT | – | – | – |
| Underfit | + L-BFGS | – | – | – |

fixed-size sketch followed by spectral recovery. A coordinate-wise PGF–ESPRIT variant may be included only as an ablation of the proposed spectral pipeline, in order to quantify the effect of the global subspace estimation and joint diagonalization.

When Gamma-regularized or Poisson–Gamma EM variants are included, they are reported separately as model-level comparators. These variants can be useful on overdispersed real count data, but they do not isolate the effect of the inference algorithm because they rely on different probabilistic assumptions or regularization criteria.

*Metrics.* On synthetic data, where the ground truth is known, we report the relative error on the rate matrix and on the mixture weights after optimal component matching. For both synthetic and real data, we report the PGF reconstruction error, the Hellinger distance between empirical and regenerated marginal distributions, a stabilized chi-square distance, runtime, and the stored representation size ($2J$ sketch scalars vs. $nd$ raw scalars). The distributional metrics are computed per coordinate and then averaged over dimensions.

*Sketch parameters.* The PGF–JESPRIT estimator is controlled by the number of sampling directions $M$, the number of shared base points (snapshots) $S$, the number of PGF samples per direction $L = N_m$, and the nominal sampling scale $\Delta$. The sketch then consists of $J = MSL$ complex PGF evaluations, stored as $2J$ real scalars. The damped construction (6)–(7) derives the effective scale $\delta_{\text{eff}} = \alpha\Delta$ and the damping radius $r$ from $\Delta$ and the data, guaranteeing $|t| \le 1$ at every query. Unless otherwise stated, these parameters are fixed across trials. We additionally report a short sensitivity study with respect to $\Delta$, since it controls the trade-off between numerical conditioning and phase wrapping: too small a value makes the exponential nodes difficult to separate, while too large a value drives $\delta_{\text{eff}}$ to be clamped by the damping and pushes the phases $\delta_{\text{eff}}\langle\lambda_k, u_m\rangle$ toward $\pm\pi$. Damping removes the magnitude blow-up of the empirical PGF that the undamped queries ($|t| > 1$) would otherwise incur.

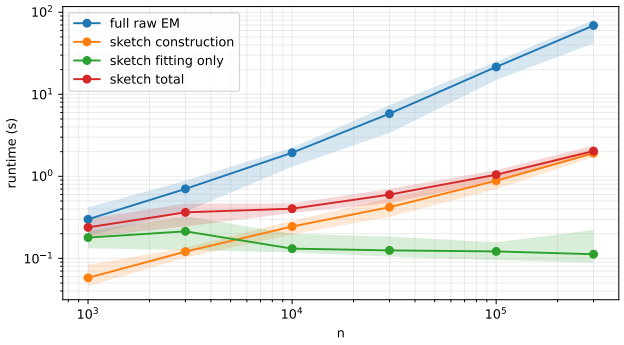



**Figure 2: Runtime vs. number of samples $n$ at a *fixed* sketch size ($M = S = L = 9$, so $J = 729$; $K = d = 3$), medians over seeds and regimes with interquartile bands, log–log. Full-data EM grows linearly in $n$, whereas the spectral fitting step (*sketch fitting only*: joint ESPRIT plus L-BFGS refinement) is essentially flat in $n$. Only the one-pass sketch construction grows with $n$, and the sketch total stays more than an order of magnitude below EM beyond $n \approx 10^4$ (about $35\times$ faster at $n = 3\times10^5$).**

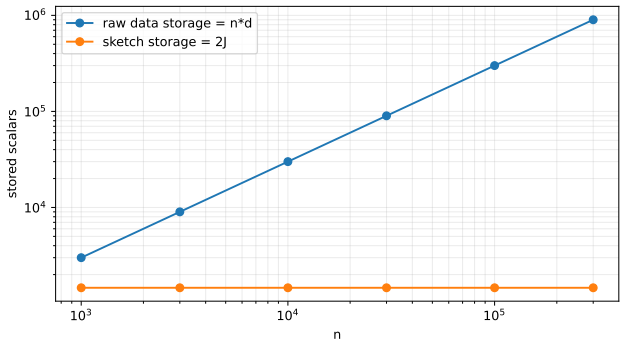



**Figure 3: Stored scalars vs. number of samples $n$, log–log, same fixed sketch size as Fig. 2. Raw storage grows as $nd$, while the sketch is a fixed $2J$ real scalars independent of $n$. At $n = 3 \times 10^5$ ($d = 3$) the sketch is more than two orders of magnitude smaller than the raw data ($2J \approx 1.5 \times 10^3$ vs. $nd \approx 9 \times 10^5$).**

*Real count data.* We include small real-data checks on the Affairs count dataset [5] and selected high-count genes from the PBMC3k single-cell RNA-seq benchmark [1]. These experiments are not intended to claim that finite Poisson mixtures are universally valid generative models for these datasets. Rather, they test whether the fitted mixtures reproduce realistic count histograms, including sparsity, imbalance, and overdispersion. On these datasets, Poisson–Gamma variants may be reported as model-level comparators, separately from the finite-Poisson algorithmic comparison.

*Scalability and storage.* Figures 2 and 3 quantify the central computational claim: at a fixed sketch size, both the memory footprint

**Table 2: Goodness-of-fit on real count datasets. Poisson–Gamma variants, when included, are model-level comparators rather than direct algorithmic baselines; "–" marks values pending the final runs. [TODO: Fill from the real-data experiments.]**

| Dataset | Method | $K$ | Hellinger | PGF err. |
|---|---|---|---|---|
| Affairs | Finite Poisson EM | – | – | – |
| Affairs | PGF–JESPRIT | – | – | – |
| Affairs | Poisson–Gamma EM | – | – | – |
| PBMC3k | Finite Poisson EM | – | – | – |
| PBMC3k | PGF–JESPRIT | – | – | – |
| PBMC3k | Poisson–Gamma EM | – | – | – |

and the recovery cost are independent of the number of observations $n$. The sketch occupies a constant $2J$ real scalars, while raw storage grows as $nd$; at $n = 3 \times 10^5$ with $d = 3$ this is a fixed $\approx 1.5 \times 10^3$ scalars against $\approx 9 \times 10^5$, more than two orders of magnitude smaller (Fig. 3). The spectral fitting step – global SVD, similarity diagonalization, NNLS, simplex weights, and the L-BFGS refinement – is flat in $n$ because it acts only on the sketch, whereas full-data EM scales linearly in $n$ per iteration. Consequently the sketch pipeline overtakes EM beyond $n \approx 10^4$ and is roughly 35× faster at $n = 3 \times 10^5$ (Fig. 2); the only data-dependent cost is the single streaming pass that builds the sketch.

*Discussion and limitations.* [TODO: Add the final accuracy findings (rate/weight/PGF error, Hellinger) once the synthetic and real tables are filled.] We expect PGF–JESPRIT to be most effective when the mixture has a moderate number of components and the rate vectors are sufficiently separated, so that the empirical PGF has a clear exponential structure and the joint subspace recovers coherent components. Several limitations remain. First, the nominal scale $\Delta$ still matters because it affects both node conditioning and phase wrapping; the damping fixes the *magnitude* instability of the undamped queries (it forces $|t| \le 1$ and matches the bias to the $n^{-1/2}$ sampling fluctuation), but it does not remove the conditioning/wrapping trade-off, which is why we report a $\Delta$ sensitivity study. Second, the spectral recovery is algebraic and does not directly optimize a fit criterion; the L-BFGS refinement of (25) mitigates this by minimizing the exact PGF loss after recovery, but in highly overlapping regimes a good PGF fit may still trail EM on histogram-based metrics. Finally, comparisons with Poisson–Gamma variants should be read as model-level comparisons, since differences may reflect both the inference procedure and the probabilistic assumptions.

## 5 Conclusion

We presented a scalable framework for mixed Poisson inference from empirical PGF sketches using joint ESPRIT. The method evaluates the empirical PGF at damped query points that stay inside the unit disk while preserving the exact sum-of-exponentials structure, recovers multidimensional rate vectors through a global subspace and a non-unitary similarity diagonalization, estimates rates by non-negative least squares and weights by simplex-constrained least squares, and finishes with a quasi-Newton refinement of the exact PGF loss using a closed-form analytic gradient. Because the sketch has a fixed size, the recovery cost and storage do not grow with the number of observations, which we confirm empirically against full-data EM. The final paper will complete the empirical study by separating algorithmic comparisons under the finite product-Poisson mixture model from model-level comparisons involving alternative count-data models. [TODO: Add final accuracy summary once the experiment tables are complete.]

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